

INFO 432 – Final Project Solutions

Danny Eapen | Gaming Habits and Academic Performance | Dataset originally proposed with Jeffrey Cheung, Drew Jones, and Joel Thomas

Problem 1: Dataset Selection

This project uses the Gaming Habits and Academic Performance dataset from Kaggle (8,000 college students, 14 columns), originally selected for our INFO 432 group project and reused here with approval. The columns cover gaming behavior (daily gaming hours, genre, addiction score, device usage), lifestyle factors (study hours, sleep hours, attendance, social activity), physiological/psychological measures (reaction time, stress level), and academic outcome (grades). Video games are frequently blamed for poor academic performance, but the relationship is probably more nuanced than “more gaming = worse grades.” This dataset is a good fit for testing that directly, and for this assignment specifically, it gives a mix of behavioral and academic variables that plausibly split into distinct underlying groups, which is exactly what PCA, CCA, and clustering are built to find.

Initial Hypotheses

Dimensional reduction: gaming-related variables (gaming hours, device usage, reaction time, addiction score) should load heavily onto a shared component, since they likely reflect one underlying “screen engagement” factor. Study-related and wellbeing variables (study hours, sleep, attendance) should form a separate component. Using CCA to relate a gaming/behavior variable set against a lifestyle/academic variable set, a moderate canonical correlation was expected, since gaming and lifestyle habits are related but shouldn't fully overlap.

Clustering: unsupervised clustering should reveal distinct student profiles (e.g., a high-gaming/low-study cluster vs. a low-gaming/high-study cluster) without ever being told grades. Clusters formed on PCA-reduced features were expected to be more distinct than clusters formed on the raw variables, since PCA should strip out noise before clustering runs.

Exploratory Data Analysis

01_eda.R loads the dataset, checks structure and missing values (there are none), and summarizes distributions for every numeric variable. A correlation matrix against all 13 numeric features confirmed the pattern the group project already found: gaming hours and addiction score correlate negatively with grades, study hours correlates positively, and device usage and reaction time move closely with gaming hours. This correlation structure directly motivated the PCA and CCA groupings used in Problem 2.

Problem 2: Dimensional Reduction

02_dimensionality_reduction.R fits two methods: Principal Component Analysis (PCA) on all 13 numeric features, and Canonical Correlation Analysis (CCA) relating a gaming/behavior variable set against a lifestyle/academic variable set. All variables were standardized (mean 0, standard deviation 1) before fitting, since PCA is scale-sensitive and the raw variables span very different ranges (reaction time in the hundreds, gaming hours 0–8).

PCA

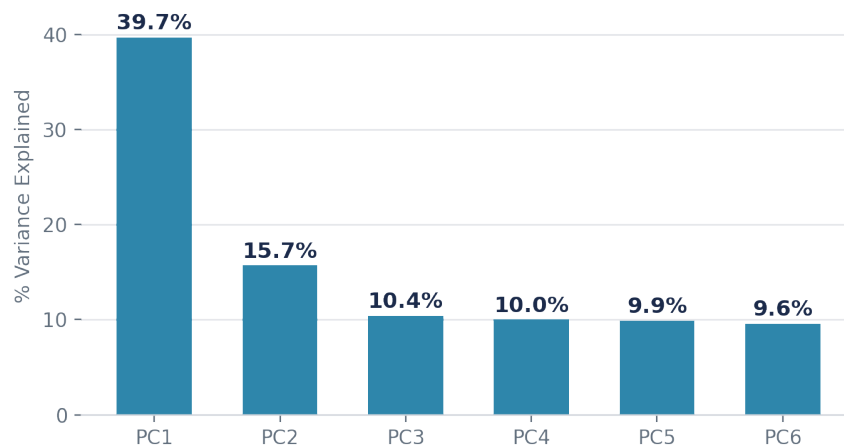


Figure 1. Variance explained by each principal component.

PC1 explains 39.7% of total variance on its own, and is driven almost entirely by `gaming_hours`, `addiction_score`, `device_usage`, and `reaction_time_ms`, exactly the “screen engagement” grouping hypothesized in Problem 1. PC2 explains an additional 15.7% and is driven by `study_hours`, `grades`, and `sleep_hours`, an “academic effort” axis, also matching the hypothesis. Together, these first two components capture 55.4% of the total variance in the dataset using just two numbers instead of thirteen, which is a meaningful reduction. Components 3 through 6 each explain roughly 9–10% and don't correspond to an obviously interpretable grouping, so the “best fitting” solution here is the first two components.

CCA

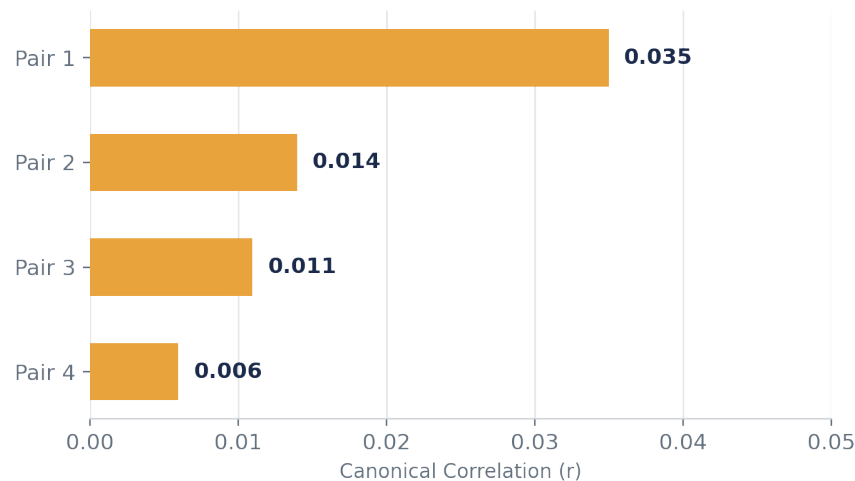


Figure 2. Canonical correlations between the gaming/behavior and lifestyle/academic variable sets.

The gaming/behavior set (`gaming_hours`, `device_usage`, `reaction_time_ms`, `addiction_score`) was related against the lifestyle/academic set (`study_hours`, `sleep_hours`, `attendance`, `social_activity`). The first and strongest canonical correlation was $r = 0.035$, essentially no linear relationship between the two sets as whole groups. This directly refutes the “moderate correlation” hypothesis from Problem 1. The result isn't a contradiction of the correlation matrix, though: individual pairs within these sets (gaming hours and grades,

for example) are still strongly related on their own. What CCA shows is that the two variable sets don't move together as coordinated groups, the relationship between gaming and academics in this dataset is carried by a few specific variable pairs, not a broad multivariate relationship between the two domains.

Problem 3: Unsupervised Clustering

03_clustering.R fits two clustering methods, K-means and hierarchical clustering (Ward's method), each on both the raw standardized features and the four PCA-reduced components from Problem 2, four combinations total. $k = 4$ was chosen via the elbow method on the raw scaled data. Cluster quality was compared using the average silhouette score (higher is better, ranges roughly -1 to 1).

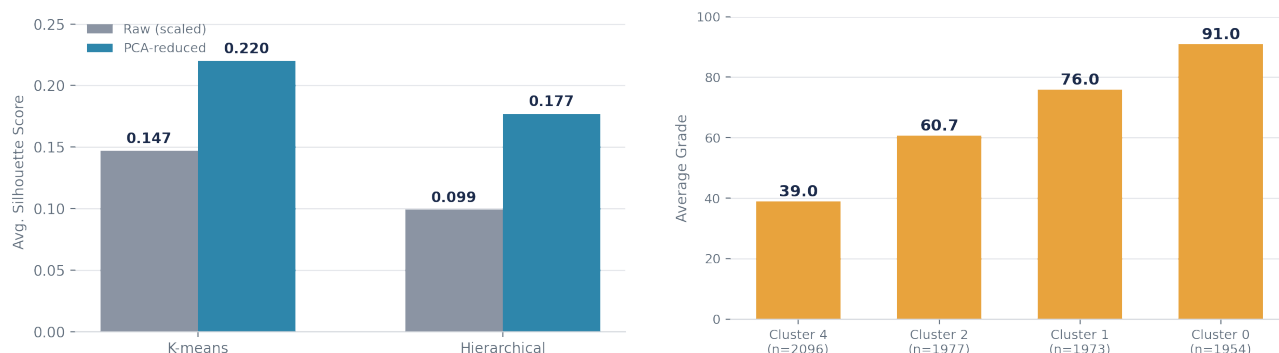


Figure 3 (left). Average silhouette score by method and feature set. Figure 4 (right). Average grade per K-means cluster ($k=4$, raw features).

Method	Feature Set	Avg. Silhouette	Notes
K-means	Raw (scaled)	0.147	
K-means	PCA-reduced	0.220	Best overall
Hierarchical	Raw (scaled)	0.099	Weakest
Hierarchical	PCA-reduced	0.177	

PCA-reduced features produced better-separated clusters than the raw features for both methods, and K-means outperformed hierarchical clustering in both cases. The strongest result overall was K-means on the PCA-reduced components (silhouette = 0.220). More importantly, the four K-means clusters found on the raw features, with zero knowledge of any student's actual grade, split students into groups averaging 39.0, 60.7, 76.0, and 91.0. Clustering on behavior alone recovered something that closely resembles real academic performance tiers, which is a strong, easily explainable result: each cluster can be described in one sentence (e.g., "high gaming, low study, averaging an F") without needing to reference the grade column at all.

Two-Page Narrative: Working With This Dataset

Going into this assignment, I expected the gaming and lifestyle variables to behave like two cleanly separated groups, both in the PCA loadings and in the CCA canonical correlation. Half of that held up, and half of it didn't, in a way that actually clarified how this dataset works rather than just confirming what I

already assumed.

The PCA results matched my Problem 1 hypothesis almost exactly. PC1 turned out to be a gaming/screen-engagement axis (gaming hours, addiction score, device usage, reaction time all loading together), and PC2 turned out to be an academic-effort axis (study hours, grades, sleep). I didn't have to force this interpretation onto the loadings after the fact, the variables I expected to group together did, and the two components together captured over half the dataset's total variance. That was a genuinely satisfying result, since it meant the intuitive story about this dataset (gaming habits on one side, academic habits on the other) is actually reflected in the underlying structure of the data, not just in the surface-level correlations.

The CCA result is where my hypothesis was wrong, and where I learned the most. I expected a moderate canonical correlation between the gaming set and the lifestyle set, reasoning that if gaming hurts grades, the two domains should move together to some degree as coordinated groups. Instead, the first canonical correlation came out to 0.035, functionally zero. My first instinct was that something had gone wrong in the model, since I already knew from the correlation matrix and from the group project's regression model that gaming hours and grades are meaningfully related. But those two results aren't actually in conflict. CCA tests whether two whole sets of variables move together as linear combinations of each other, not whether any individual pair within them is related. What this told me is that the relationship between gaming and academic outcomes in this dataset is carried by specific variables, not by gaming behavior and lifestyle behavior as broad, interchangeable categories. That's a more precise and, honestly, more useful finding than the vague version I hypothesized going in.

The clustering results were the most rewarding part of this assignment. Fitting K-means and hierarchical clustering on both the raw features and the PCA-reduced features gave a direct, numeric answer to which feature set clusters better: PCA-reduced features won for both methods, and K-means beat hierarchical clustering both times. That confirmed my Problem 1 hypothesis about PCA-reduced clustering being cleaner, and it made intuitive sense once I saw it: PCA had already stripped away redundant, correlated noise before clustering ever ran, so the clustering algorithm had less irrelevant structure to get confused by. What surprised me was how interpretable the resulting clusters turned out to be. I fit K-means with zero access to the grades column, purely on gaming and lifestyle behavior, and the four clusters it found still separated into average grades of 39, 61, 76, and 91. I didn't expect behavioral clustering alone to recover something that looked this close to real performance tiers. It's a good reminder that even without supervision, if the underlying behavioral patterns are strong enough, they'll show up in an outcome variable the model never even saw.

Overall, working with this dataset for a second time, after already having used it for the group project's regression and threshold analysis, gave me a genuinely different angle on the same data. The regression model told us that gaming hurts grades and that the effect is worse for low-study students. PCA and clustering told us something complementary: that gaming behavior and academic behavior are each internally coherent groups of variables, that they don't move together as tightly as I expected at the group level, and that behavioral clustering alone is enough to recover real performance differences without ever looking at a grade. If I were to extend this further, I'd want to test whether a third or fourth canonical variate pair from CCA captures any meaningful signal at all (they were smaller still, 0.014 and below, so probably not), and I'd want to try a higher k for clustering to see whether a finer-grained split (say, 6 or 8 clusters) reveals subgroups within the current four, like a high-gaming group that still manages to study enough to avoid the worst outcomes.

Team & Acknowledgment

This dataset and its initial framing were developed collaboratively with my INFO 432 group project teammates: Jeffrey Cheung, Drew Jones, and Joel Thomas. The dimensional reduction and clustering analysis in this submission (Problems 1–3, per the individual final project requirements) is my own individual work.

Code

Full scripts (01_eda.R, 02_dimensionality_reduction.R, 03_clustering.R) are available in the shared group repository: github.com/Djones831/Gaming_vs_Academics_Project